HW 3 - Aquaculture Analysis

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Prepare data

```
# Load packages and data

library(sf)
library(tmap)
library(tidyverse)
library(spData)
library(knitr)
library(gt)
library(kableExtra) # table formatting
library(spDataLarge) # spatial data
library(geodata) # spatial data
library(terra)
library(here)
set_theme(theme_bw())
```

```
# Load data

eez <- read_sf(here("data", "wc_regions_clean.shp"))

depth <- rast(here("data", "depth.tif"))


# stack the rasters in the call for the data
sst <- rast(list.files(here("data"),
pattern = "average_annual_sst", full.names = TRUE))
```

```r
# use project because sst is a raster to make the CRSs match
sst <- project(sst, crs(depth))
eez <- st_transform(eez, crs(depth))

# Check crs
crs(eez) == crs(depth)
```

```
[1] TRUE
```

```r
crs(sst) == crs(depth)
```

```
[1] TRUE
```

Process data

```r
# take mean temps
mean_sst <- mean(sst, na.rm = TRUE)
sst_c <- mean_sst - 273.15
```

```r
# crop rasters to match extents
depth_crop <- crop(depth, sst_c)
```

```r
# resolutions match using bilinear resampling method
depth_resampled <- resample(depth_crop, sst_c, method = "bilinear")

cat("After resampling:\n")
```

```
After resampling:
```

```r
cat("Resolutions match:", all(res(depth_resampled) == res(sst_c)), "\n")
```

```
Resolutions match: TRUE
```

```r
cat(
  "Origins match:    ",
  all(round(origin(depth_resampled), 8) == round(origin(sst_c), 8)),
  "\n"
)
```

```
Origins match:     TRUE
```

```
cat("Extents match:    ", ext(depth_resampled) == ext(sst_c), "\n")
```

```
Extents match:     TRUE
```

```
# reclassify for suitable area
# define a reclassification matrix
rcl_dep <- matrix(
  c(
    -Inf,
    -70,
    0,
    -70,
    0,
    1,
    0,
    Inf,
    0
  ),
  ncol = 3,
  byrow = TRUE
)

# Apply the matrix to reclassify the raster, making all cells 0 or 1
depth_rcl <- terra::classify(depth_resampled, rcl = rcl_dep)
```

```
# reclassify for suitable area
# Define a reclassification matrix
rcl_sst <- matrix(c(-Inf, 11, 0, 11, 30, 1, 30, Inf, 0),
 ncol = 3, byrow = TRUE)

# Apply the matrix to reclassify the raster, making all cells 0 or 1
sst_rcl <- terra::classify(sst_c, rcl = rcl_sst)

# combine the two reclassified rasters
combined <- c(sst_rcl, depth_rcl)
```

Find suitable locations

```
# find suitable locations with lapp
# multiply both of the reclassified

# Stack them, then apply a multiplication
# function of the two rasters as arguments
suitable <- lapp(combined, fun = function(sst_rcl, depth_rcl) {
  sst_rcl * depth_rcl
})

# rasterize the eez data and make sure the
# dimensions/extents are correct using mask()
eez_rast <- rasterize(eez, suitable, field = "rgn_id")
suitable <- mask(suitable, eez_rast)

# find the physical areas in km using cellSize()
suitable_areas <- cellSize(suitable, unit = "km", mask = TRUE)

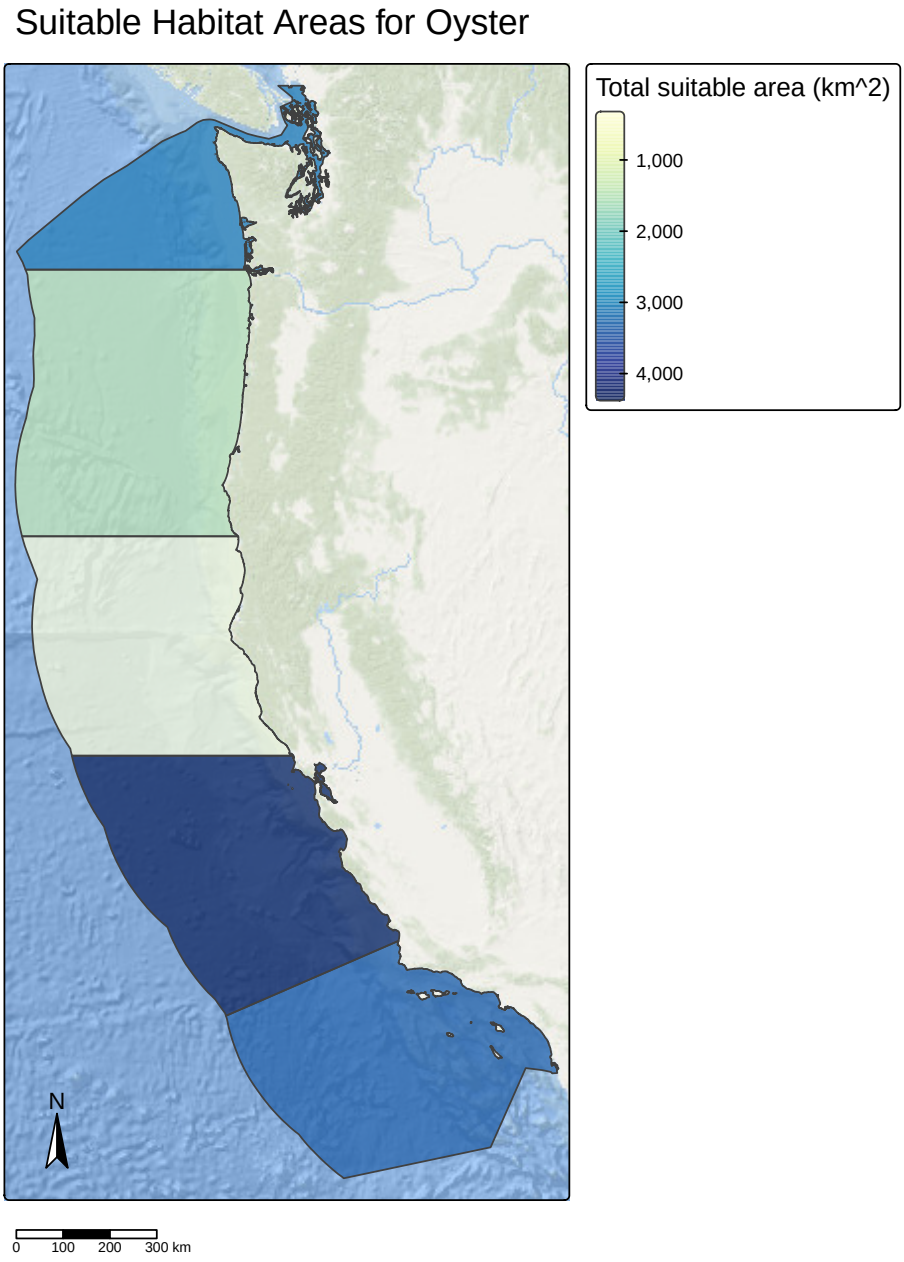
# make sure dims and exts are correct
suitable_zone_mask <- mask(suitable_areas, suitable, maskvalues = 0)

# sum the total suitable areas using zonal()
zone <- zonal(suitable_zone_mask, eez_rast, fun = "sum", na.rm = TRUE)
```

```
# Join zones with suitable areas to the eez
#  outlines by region id so we know where has the most suitable habitat area
eez_zones <- left_join(eez, zone, by = "rgn_id") |>
  arrange(desc(area)) # largest suitable area first

# map it
tm_basemap("Esri.OceanBasemap") + # basemap
  tm_shape(eez_zones) + # outline the economic zones
  tm_polygons(
    fill = "area",
    fill_alpha = 0.7,
    fill.scale = tm_scale_continuous(values = "brewer.yl_gn_bu"),
    # fill by suitable area scale
    fill.legend = tm_legend(
      title = "Total suitable area (km^2)"
    )
  ) +
```

```
  tm_title("Suitable Habitat Areas for Oyster", size = 1.25) +
  tm_scalebar(
    position = tm_pos_out(
      "center",
      "bottom"
    )
  ) +
  tm_compass(
    position = tm_pos_in(
      "left",
      "bottom"
    )
  )
```



Interpretation:

In this map we can see the suitable habitat areas for the oyster, which live in zones with depths ranging 0-70 m below sea level and temperatures ranging 11-30º C (SeaLifeBase). In the map we

can see that the region with the most suitable habitat area is in the Southern California region, where suitable area is around 4,000 km². This is likely due to more suitable temperatures in warmer conditions in Southern California, as compared to Northern California.

Determine the most suitable EEZ

```
# make a table for the data
eez_zones |>
  st_drop_geometry() |>
  select(rgn_id, area, area_km2, rgn) |>
  arrange(desc(area)) |>
  mutate(pct_suitable = area / area_km2 * 100) |>
  kable(
    digits = 1,
    col.names = c(
      "EEZ Region",
      "Suitable Area (km²)",
      "Total EEZ Area (km²)",
      "Suitable (%)",
      "Region Name"
    ),
    caption = "Suitable oyster aquaculture area by West Coast EEZ",
    booktabs = TRUE
  ) |>
  kable_styling(latex_options = c("striped", "hold_position"))
```

Table 1: Suitable oyster aquaculture area by West Coast EEZ

EEZ Region	Suitable Area (km²)	Total EEZ Area (km²)	Suitable (%)	Region Name
3	4389.6	202738.3	Central California	2.2
4	3347.8	206860.8	Southern California	1.6
5	3062.6	66898.3	Washington	4.6
1	1380.6	179994.1	Oregon	0.8
2	340.3	164378.8	Northern California	0.2

Make a function to recreate the analysis on other species

::: {.cell}

```r
suitable_eez_map <- function(
 species_name, min_temp, max_temp, min_depth, max_depth)
 {
 # Validate inputs
 if (min_temp >= max_temp) stop("min_temp must be less than max_temp")
 if (min_depth >= max_depth) stop("min_depth must be less than max_depth")
 if (min_depth < 0) stop("Enter depths as positive meters below sea level")

 # Reclassify SST
 rcl_temp <- matrix(c(-Inf, min_temp, 0,
                min_temp, max_temp, 1,
                max_temp, Inf, 0),
              ncol = 3, byrow = TRUE)
              # Apply the matrix to reclassify the raster,
              # making all cells 0 or 1
temp_rcl <- terra::classify(sst_c, rcl = rcl_temp)

 # Reclassify depth
 rcl_bathym <- matrix(c(-Inf, -max_depth, 0,
                      -max_depth, -min_depth, 1,
                      -min_depth,  Inf, 0),
                    ncol = 3, byrow = TRUE)
              # Apply the matrix to reclassify the raster,
              # making all cells 0 or 1
bathym_rcl <- terra::classify(depth_resampled, rcl = rcl_bathym)

 # Combine conditions
depth_sst <- c(bathym_rcl, temp_rcl)

 # Suitable area per EEZ
 # mask to eez_rast, cellSize(), mask out 0s, zonal(sum)
 suitable <- lapp(depth_sst, fun = function(temp_rcl, bathym_rcl) {
 temp_rcl * bathym_rcl
})

# rasterize the eez data and mask along with suitable areas
eez_rast <- rasterize(eez, suitable, field = "rgn_id")
suitable <- mask(suitable, eez_rast)

suitable_areas <- cellSize(suitable, unit = "km", mask = TRUE)
suitable_zone_mask <- mask(suitable_areas, suitable, maskvalues = 0)
```

```
zone <- zonal(suitable_zone_mask, eez_rast, fun = "sum", na.rm = TRUE)

 # Join and map
eez_zones <- left_join(eez, zone, by = "rgn_id") |>
 arrange(desc(area)) # largest suitable area first

map <- tm_basemap("Esri.OceanBasemap") +
 tm_shape(eez_zones) +
 tm_polygons(
   fill = "area",
   fill_alpha = 0.7,
   fill.scale = tm_scale_continuous(values = "brewer.yl_gn_bu"),
   fill.legend = tm_legend(
     title = "Total suitable area (km^2)"
   )
 ) +
 tm_title(paste("Suitable Habitat Areas for ", species_name), size = 1.25) +
 tm_scalebar(
   position = tm_pos_out(
     "center",
     "bottom"
   )
 ) +
 tm_compass(
   position = tm_pos_in(
     "left",
     "bottom"
   )
 )

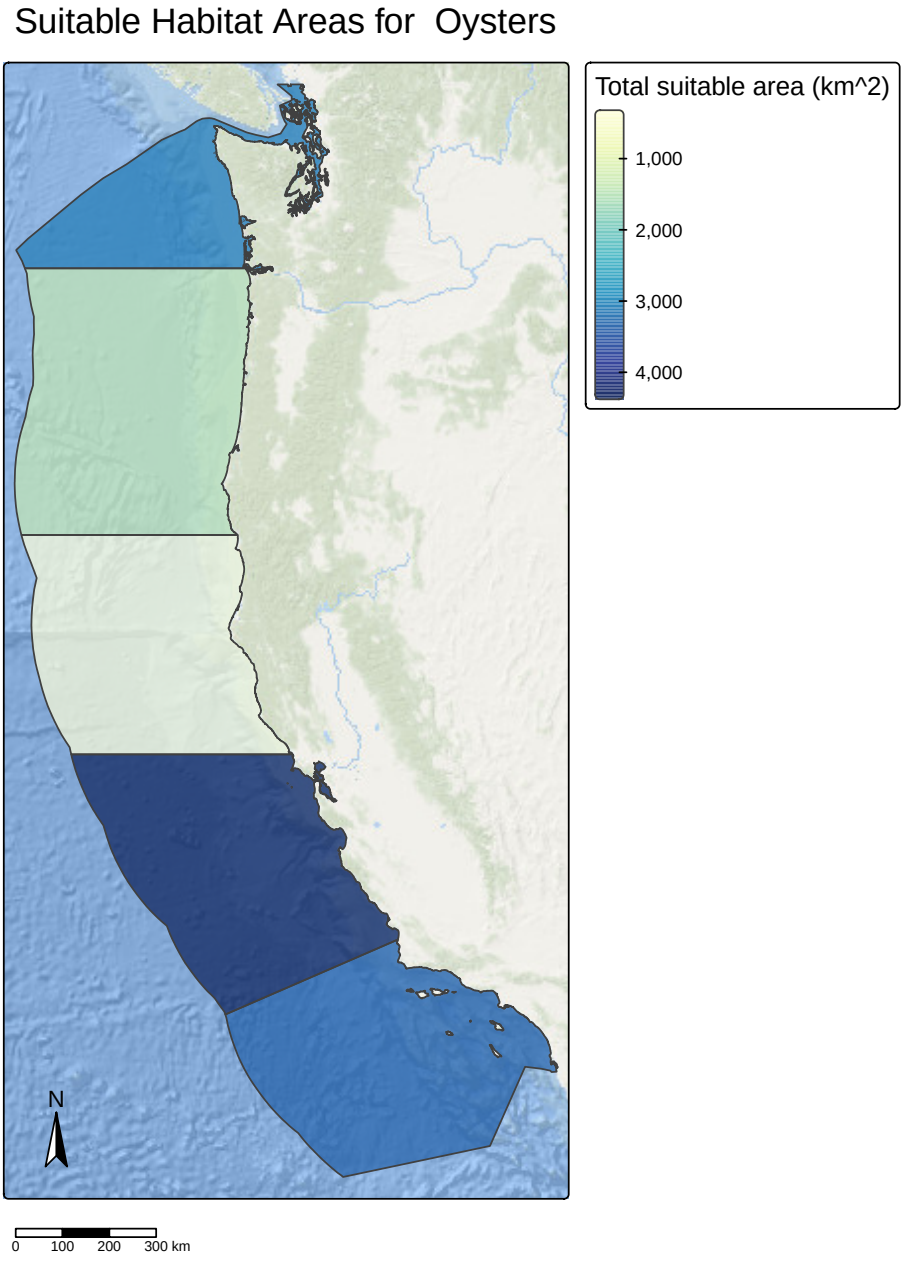
 # Return
return(map)
}
```

:::

Test the function on oysters

::: {.cell}

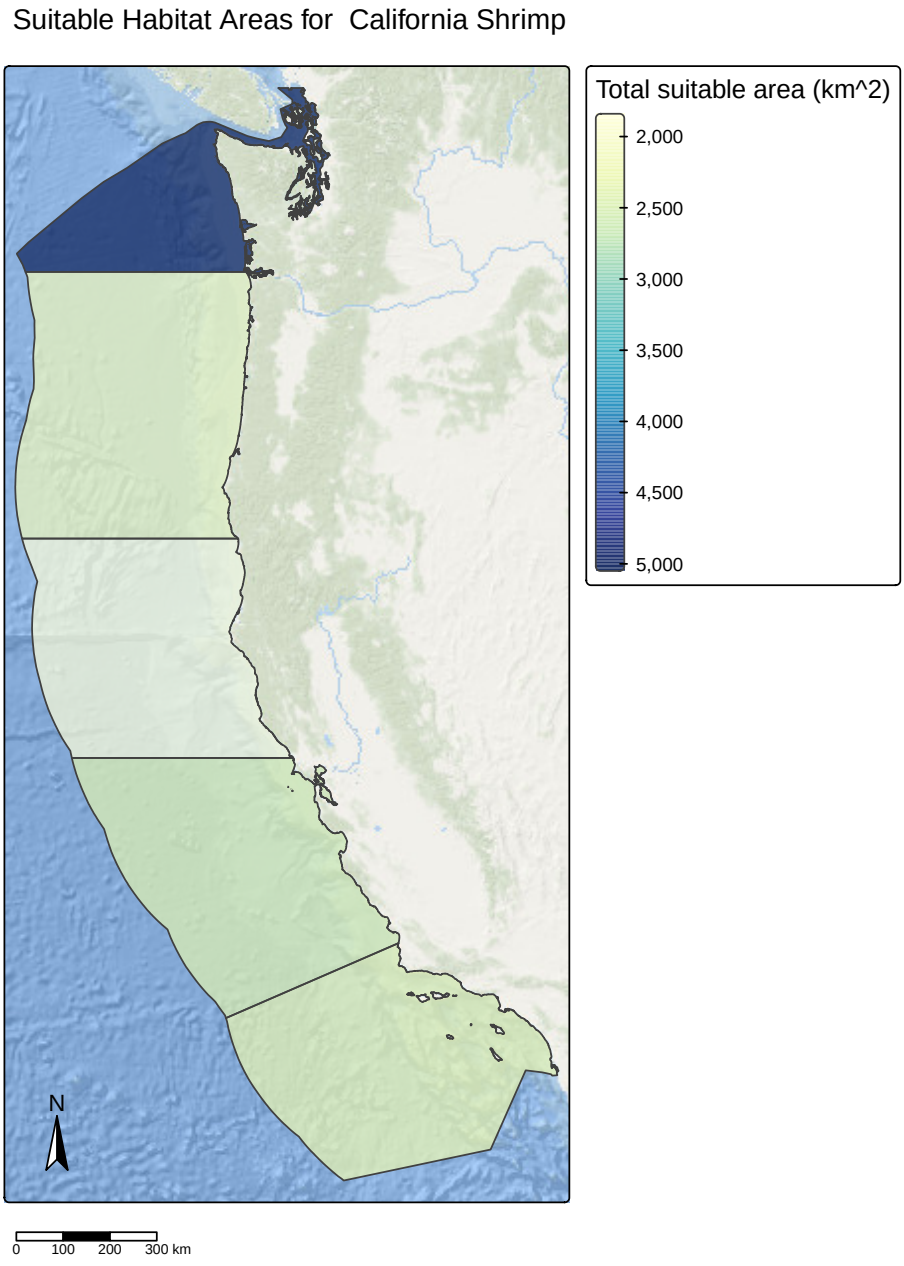
```
suitable_eez_map("Oysters", 11, 30, 0, 70)
```



::: {.cell-output-display}
::: :::

Test the function on shrimp

```
suitable_eez_map("California Shrimp", 8, 20, 5, 50)
```



Interpretation:

In this map we can see the suitable habitat areas for the California Shrimp, which live in zones with depths ranging 5-50 m below sea level and temperatures ranging 8-20º C (SeaLifeBase). In the map we can see that the region with the most suitable habitat area is in the Northern California region, where suitable area is above 4,000 km². This is likely due to more suitable temperatures Northern California, as compared to Southern California.

References

"Search SeaLifeBase." Www.Sealifebase.Ca, https://www.sealifebase.ca. Accessed 5 Nov. 2026.

"NOAA Coral Reef Watch Daily Global 5Km Satellite Coral Bleaching Heat Stress SST Anomaly Product (Version 3.1)." Coralreefwatch.Noaa.Gov, https://coralreefwatch.noaa.gov/product/5km/index_5km_ssta.php. Accessed 5 Nov. 2026.

"Gridded Bathymetry Data (General Bathymetric Chart of the Oceans).." Gebco, https://www.gebco.net/data_and_products/gridded_bathymetry_data/. Accessed 5 Nov. 2026.

"Marine Regions." Www.Marineregions.Org, https://www.marineregions.org/eez.php. Accessed 5 Nov. 2026.